

Testing Report of the Prima and Optiprofiler Packages

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May 19, 2026 5:30pm +08:00

摘要

This is the testing report of the Prima and Optiprofiler packages. Firstly, We try to use Matlab Online(in Mathworks website, offering Linux environment) to test the two packages and complete the tasks, saving a great amount of time installing both the packages themselves and the prerequisites. However, due to the absence of parallel computing environment(discussed in details below), we are unable to complete the second task online, forcing us to turn back to local environment.

1 The Prima Package

1.1 Installation

Download the Prima package from <https://github.com/libprima/prima>. Prerequisites: Matlab Mex run `'setup'` in the `'prima'` folder to install the package. Then run `'options.debug = true; options.single = true; options.quadruple = true; setup(options); testprima'` to test the package.

1.2 Testing

Run `'testprima_ex'` in the `'prima/matlab/tests'` folder to test the package.

Issues:

- (a) The `'testprima_ex'` will run the `'setup'` function, which has already been done.

1.3 The Rosenbrock Function

Use `prima` to solve the following optimization problem in Matlab. One can refer to `'help prima'` and the example in `'prima/matlab/examples'`. **Problem:** Object function: the Rosenbrock function in $\mathbb{R}^n$. Initial point: $(-1, -1, \dots, -1) \in \mathbb{R}^n$. Testing in different constraints:

(a) unconstrained

(b) Boundary constrained: $x \leq 0$.

(c) linear constrained: $\sum x \leq 1, x \geq 0$.

(d) nonlinear constrained: $\sum x^2 \leq 1, x \geq 0$.

Answer From the Example: Run `“testprimae”`, we get the result in `ExampleOutput.txt`■

Testing Output: Run the code in `PrimaTesting.m`, we get the result in `TestingOutput.txt`.

Note that we have obtained different results of the same constraint, indicating that the choice of the initial point matters when the global minimum doesn't exist. If we replace the initial point with `0;0;0` in the testing code, we will obtain the same result as in the example code.

2 The Optiprofiler Package

2.1 Installation

Use `git` to clone the repository. More specifically, run `'git clone https://github.com/optiprofiler/optiprofiler'` repository is clone into `'optiprofiler'`. Then run `'cd optiprofiler'`, followed by `'ls'` to check if the copying process is successful. I forget where exactly the path in my local computer is, so I run `'pwd'` to find the answer. Open the file you have cloned the repository to. Run `'setup'` in the `'optiprofiler'` folder to install the package.

2.2 Testing

Run `'example 1; example 2; example 3; example 4;'` in the `'optiprofiler/matlab/examples'` folder to test the package.■

47 2.3 Testint Prima

48 Complete to following two Matlab Testing tasks. One can refer to 'help benchmark'
49 and the examples in <https://www.optprof.com/latest/user/usage.html>.

$\mathbb{T} \sim \approx \not\sim$: Solver 1: Prima, with precision 'double'. Solver 2: Prima, with precision 'single'. Testing Problem Type(ptype): 'ubln'(testing unconstrained, bound constrained, linear constrained, and nonlinear constrained problems altogether). Testing Problem Dimension(mindim, maxdim): 2 20. Testing Problem Feature(feature_{name}) : '*plain'and'noisy'(testingseperatel*

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